# MyDesign Portfolio

## *Element H*

## Element H:  Prototype Testing and Data Collection Plan

**Testing Details** *(H1, H2, H3, H4)*

| | requirement | testing setup and procedure | test cases and number of trials | expected range of results | pass/fail criteria, goal |
|---|---|---|---|---|---|
| 1 | The mechanism must clamp to the pole | 1: Take open end of clamp and bring it toward the light pole<br>2: Fasten the clamp to pole | Fasten and unfasten clamp to pole<br>(3 trials) | Clamp May:<br>1: stand strong<br>2: Fall downward | Clamp must hold onto pole |
| 2 | The mechanism must be sturdy. | 1:Clamp the mechanism to the pole<br>2:attach board to the holder<br>3: apply pressure and see if the clamp is stable. | Fasten and then apply pressure<br>(3 trials) | Clamp may:<br>1: bend in a different direction<br>2: Remain in the same position sturdy | Clamp must be sturdy enough to not bend or collapse |
| 3 | The mechanism must be adjustable | 1: attach clamp to finger mold and pole<br>2:move clamp in different directions (left, right, up, down) | Push clamp one by one in all 4 different directions<br>(3 trials) | Clamp may:<br>1:Move in the direction desired<br>2: Clamp may "resist" and not move | Clamp must be able to adjust in different directions |
| 4 | The mechanism must be cost effective | 1:Review the total costs<br>2: determine if it is under the budget | Review costs and see if it fits under budget<br>(1 trial) | Clamp May:<br>1: cost more than 50$<br>2:cost less than 50$ | Clamp mechanism must cost under 50$ |



Element H continues on the next page →

##### Feedback From Field Experts *(H5)*

a) You must provide evidence in this section that your testing plan has been reviewed by 2-3 field experts.

Dr. Jennifer Love (Northeastern University professor) reviewed this test plan on Sunday April 27, 2025. It seems to me that the team's functions, objectives and constraints in their project's original Design Brief should be used to formulate a test plan. For example, the objective "Sturdy Construction - The system should be capable of supporting a reflective whiteboard up to 230 grams without compromising its stability" should be used to write a test plan requirement that directly assesses whether the system can support a whiteboard of 230 grams. You have a good start with 4 test requirements but your 2nd test requirement about sturdiness could be reworded to incorporate the "up to 230 grams" phrase instead of applying a "pressure". You're on the right track!

| | | | | | |
|---|---|---|---|---|---|
| 2 | The mechanism must be sturdy. | 1:Clamp the mechanism to the pole<br>2:attach board to the holder<br>3: apply pressure and see if the clamp is stable. | Fasten and then apply pressure<br>(3 trials) | Clamp may:<br>1: bend in a different direction<br>2: Remain in the same position sturdy | Clamp must be sturdy enough to nor bend or change. |

##### Revised Testing Details That Incorporate Field Expert Feedback *(H5)*

| | requirement | testing setup and procedure | test cases and number of trials | expected range of results | pass/fail criteria, goal |
|---|---|---|---|---|---|
| 1 (highest priority) | The mechanism must clamp to the pole | 1: Take open end of clamp and bring it toward the light pole<br>2: Fasten the clamp to pole | Fasten and unfasten clamp to pole<br>(3 trials) | Clamp May:<br>1: stand strong<br>2: Fall downward | Clamp must hold onto pole |
| 2 | The mechanism must be sturdy. | 1:Clamp the mechanism to the pole<br>2:attach board to the holder<br>3: apply pressure and see if the clamp is stable. | Fasten and then apply pressure<br>(3 trials) | Clamp may:<br>1: bend in a different direction<br>2: Remain in the same position sturdy | Clamp must be sturdy enough to nor bend or change with a weight up to 230 grams |
| 3 | The mechanism | 1: attach clamp to finger | Push clamp one | Clamp may: | Clamp must be |

| | | | | | |
|---|---|---|---|---|---|
| | must be adjustable | mold and pole<br>2:move clamp in different directions (left, right, up, down) | by one in all 4 different directions<br>(3 trials) | 1:Move in the direction desired<br>2: Clamp may "resist" and not move | able to adjust in different directions |
| 4 (lowest Priority) | The mechanism must be cost effective | 1:Review the total costs<br>2: determine if it is under the budget | Review costs and see if it fits under budget<br>(1 trial) | Clamp May:<br>1: cost more than 50$<br>2:cost less than 50$ | Clamp mechanism must cost under 50$ |